Effectiveness of Artificial Intelligence in Technical Writing Skills: Academic Performance of Second-year Bachelor of Science in Biology Students at Aklan State University

Abstract

This study, titled "Effectiveness of Artificial Intelligence in Technical Writing Skills: Academic Performance of Second-Year Bachelor of Science in Biology Students at Aklan State University," aims to examine the impact of artificial intelligence (AI) tools on the technical writing abilities of undergraduate biology students. As AI-powered tools such as grammar checkers, paraphrasers, and content generators become increasingly integrated into academic work, this research seeks to determine whether their use enhances or hinders students' writing performance and cognitive skill development. The study employs a quantitative descriptive-correlational research design to analyze the relationship between the level of AI tool usage and students' academic performance in technical writing. It focuses on key aspects such as clarity, coherence, conciseness, accuracy, and audience appropriateness in scientific writing. Data will be collected through structured survey questionnaires administered to second-year Bachelor of Science in Biology students at Aklan State University. Furthermore, the research explores both the advantages and potential drawbacks of AI integration in academic writing, including concerns related to dependency, academic integrity, and critical thinking development. The findings of this study aim to provide valuable insights for students, educators, and academic institutions in developing effective strategies and guidelines for the responsible and balanced use of AI in technical writing. Ultimately, this research contributes to the growing body of knowledge on the role of artificial intelligence in education and supports the improvement of teaching methodologies, curriculum development, and student learning outcomes in the field of biological sciences.